

Edit-JEPA: Latent World Models over Text Buffers

Latent planning over mutable text buffers

TextJEPA project

July 13, 2026 (repo: `TextJEPA/`, runs: `runs/edit_*`)

Idea: the world is a mutable draft

- ▶ State = a **text buffer**: a draft solution to an iGSM-style problem (same generator as the discourse track; see companion deck for the problem sampler).
- ▶ Action = a **span edit** with an intent phrase that never leaks outcomes: delete step 3 . / insert green lamps at the start . / replace step 2 : recompute dark cups .
- ▶ Corruptions have **propagated consequences**: corrupting one step's value rewrites every downstream step *consistently* (change an assumption $\rightarrow$ later derivations change). One root cause $\Rightarrow$ several defects.
- ▶ Defect definition (symbolic, order-insensitive): a buffer step whose sentence $\neq$ the true step sentence, a missing necessary step, or an extra distractor step. **Solved** = zero defects.
- ▶ Generation = **edit until perfect**: latent search over candidate edits, execute best, re-encode, repeat.

Corruption model (exact) and a verbatim sample

From the perfect draft (necessary steps, topological order), sample independently: $k_{\text{wrong}} \sim \mathcal{U}\{0..2\}$ internal steps get a wrong stated value (propagated to descendants via the corrupted value map); $k_{\text{miss}} \sim \mathcal{U}\{0, 1\}$ necessary steps removed; $k_{\text{extra}} \sim \mathcal{U}\{0, 1\}$ distractor steps inserted; re-corrupt if defect count = 0. Repair trajectories: uniformly random fixing edit per step; with $p=0.1$ (0.25 in valgrad/anchor; max 1–2) a **vandal** edit (harmful distractor insert) $\Rightarrow$ the value head sees edits that *increase* distance. **Corrupted draft** (3 defects; step 3's wrong value propagates into step 4):

[1] so the number of large shells is 11 .

[2] so the number of silver cards is 17 .

[3] so the number of green books is 11 minus 17 = 5 . $\leftarrow$ wrong (true: 17)

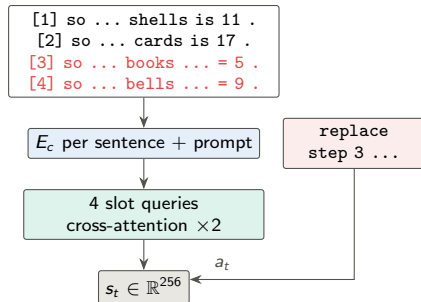
[4] so the number of golden bells is 5 times 17 = 9 . $\leftarrow$ consistent-but-wrong

missing: the step for green lamps

Fixing intents: replace step 3 : recompute green books . insert green lamps at the start .

Architecture: slot-based buffer encoder + shared JEPA core

- ▶ Chunk encoder E_c (identical to discourse track, 256-d).
- ▶ **SlotBufferEncoder**: 4 learned slot queries cross-attend over prompt and position-encoded buffer sentences; four flattened slots form the 256-d state.
- ▶ The action encoder, latent predictor F , displacement decoder, hierarchical objective, and value head are shared with the discourse architecture.
- ▶ Value target = **defects remaining** (distance to perfect).
- ▶ **Fixed outcome target**: predict a frozen random-feature embedding of the post-edit buffer from (s_t, a_t) .
- ▶ **Sentence-attentive transition**: action-conditioned queries attend to individual buffer sentences before next-state prediction, avoiding a purely pooled transition bottleneck.



Objectives, training, planning

Losses: identical composite to the discourse deck (weights 1.0 pred / 1.0 rollout / 0.5 hierarchy / 1.0 VICReg / 2.0 Delta-JEPA / 1.0 value; **anchor** adds $\mathcal{L}_{\text{chunk}}$ w=2.0 on frozen buffer embeddings).

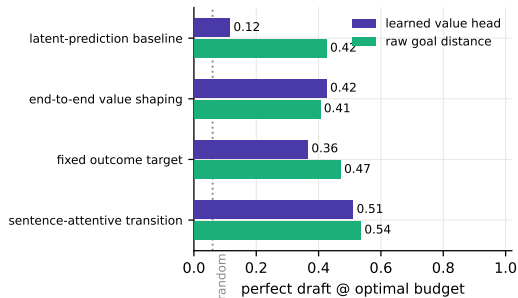
Labels: $\text{op} \in \{\text{delete, insert, replace}\}$; “was the edit a fix or vandalism” plays the goal-relevance role.

Training: batch 64 (each item encodes all $T+1$ buffer snapshots), 20 epochs over a deterministic 60k-trajectory corpus, otherwise identical protocol (AdamW 3×10^{-4} , EMA $0.99 \rightarrow 0.999$, RTX 6000, $\sim 70\text{--}90$ min).

Planning: candidates from the edit interface = $\{\text{delete } p, \text{replace } p\}$ for every position + insert for every missing necessary step + 2 harmful distractor inserts (planner must *reject* them); score $V(F(s, a_i), s_{\text{init}})$; apply argmin in the environment; re-encode; budget = initial defects + slack. Success = **perfect draft within the optimal number of edits.**

Result — the geometry was never the problem

Claim: The buffer encoder's *raw* goal-distance geometry beats every learned edit value head — the published weakness was a value-head failure.



- ▶ Value heads: latent-prediction baseline 0.115 (random 0.06), end-to-end value shaping 0.425, fixed outcome target 0.365 — all **slack-insensitive** (ties on no-op edits waste budget).
- ▶ Raw goal distance $\|\text{LN}(\hat{s}) - \text{LN}(s_{\text{goal}})\|_1$ (s_{goal} = encoded perfect buffer): latent-prediction baseline **0.425**, fixed outcome target **0.47** — $4\times$ the base value head, no value supervision used at plan time.
- ▶ Geometry regularizers (straighten/monotonicity, cf. discourse deck) change nothing here: 0.445–0.485 — $\text{corr}(d^{\text{goal}}, \text{defects})$ was already ≈ 0.75 .

Result 2 — preference learning and transition architecture

Claim: Counterfactual preference learning makes planning slack-sensitive; sentence-level attention then improves transition matching and planning.

run	value energy		goal dist.	value τ	matching
	@opt	@slack2	@opt	(audit)	(chance .10)
fixed outcome target	0.365	0.375	0.47	0.39	0.47
counterfactual preference ($K = 4$)	0.46	0.505	0.60	0.59	0.43
attention transition model	0.51	0.55	0.535	0.61	0.50
geometry-distilled value	0.42	0.42	0.55	0.42	0.45
without harmful-edit negatives	0.22	0.225	0.445	-0.07	0.39
five-objective model (no defect counts)	0.455	0.485	0.545	—	—

Readings. Preference learning gives the first slack-sensitive edit planner. Action-conditioned attention over individual sentences raises the learned-value planner from 0.46/0.505 to 0.51/0.55 and matching from 0.43 to 0.50. Geometry-distilled V matches scalar defect-count supervision (0.42 vs 0.425), while removing harmful-edit negatives makes the energy anti-correlated ($\tau = -0.07$). Matching remains below the discourse track's 1.00, so transition fidelity still limits the edit model.

Probe summary (edit track)

probe (linear, validation split)	baseline	value shaping	random encoder
edit op from Δs	1.00	1.00	1.00*
fix-vs-vandal from Δs	0.96	0.98	0.92
defects remaining from s_t	0.46	—	0.38
buffer length from s_t	0.91	—	0.73
claimed answer from s_t	0.23	—	0.28
true answer from terminal state	0.05	—	0.13

*intent phrases are trivially separable by op — the informative row is fix-vs-vandal. The two bottom rows show the **collusion signature** (trained < random control): outcome content is being smoothed away, as in the discourse track before adding a fixed outcome target. The corresponding edit model (weight 2.0) trains $\mathcal{L}_{\text{chunk}}$ to 0.005 and lifts value-head planning to 0.365, but the linear *defect-count* probe stays flat — the goal signal lives in the *metric* (see previous slide), not in a linearly decodable counter.

Diagnosis & next steps

- ▶ **Why the base value head fails:** repairs are near-deterministic given the defect set, so state prediction is easy without a linear defect counter; but the *metric* structure (distance to the perfect buffer) is there from the start — use the goal-distance energy, or distill it into V .
- ▶ **Deployable goal states:** the oracle goal uses the true buffer; a deployable version needs a goal *description* encoder (“all steps correct”) or a self-predicted fixed point $F(s, a_{\text{stop}}) \approx s$.
- ▶ **Energy ties:** counterfactual preferences make success slack-sensitive. Sentence-level attention raises matching 0.43→0.50 and planning 0.46/0.505→0.51/0.55, but the remaining gap to discourse matching indicates that transition fidelity still binds.
- ▶ **Harder corruption:** multiple root causes, order violations, subtle wrong-parent corruptions (currently only wrong values / missing / extra).
- ▶ **Long-term:** draft-level macro edits (rewrite region), and the hierarchical planner shared with the discourse track.